

THE STRUCTURE OF MATHEMATICAL REALITY

Phase III — Chapter 9

Topology: Structure Without Measurement

Quiz

Self-Study Curriculum

Before you begin

This quiz tests application, not memorisation. Answer in your own words.

Do not quote the lecture back at yourself. If you find yourself writing a phrase you remember reading, stop and rephrase it from your own understanding. A short, precise answer in your own language is worth more than a long transcription.

1. A topologist and a geometer look at the same surface. What does the topologist see that the geometer also sees? What does the geometer see that the topologist ignores?

2. You are told that a dynamical system's phase space is compact. What can you immediately infer about the system's long-term behavior?

3. Two clustering algorithms give different results on the same data. Could a topological analysis help you decide which is more "correct"? How?

4. The parameter space of a certain model has a “hole” in it, a region where the model is undefined or ill-conditioned. What is the topological significance of this hole? What practical problems does it cause?

After the quiz

For each answer, ask yourself: “*Did I explain the **why**, or did I just state the **what**?*” If you stated a fact without explanation, go back and add the reasoning.

Date: _____ **Questions where I want to go deeper:**
